



# Electric vehicle and solar energy pilot: Opportunity to address suburban energy challenges

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## ABSTRACT

Low- and moderate-income communities in suburban areas are dependent on transportation to meet basic needs. This case study describes an innovative approach by GRID Alternatives and Valley Clean Air Now (Valley CAN) to combine solar rooftop with electric vehicle incentives. Wider deployment of these technologies could decrease costs and increase access to renewable energy and EVs for potentially vulnerable families with high transportation and electricity costs.

## 1. Introduction

### 1.1. Case study introduction

The Smiths<sup>1</sup> moved to the suburbs of the San Joaquin Valley in Merced County, Calif. around 2008. During the Great Recession, the family was able to use their savings to purchase a two-bedroom home in the suburbs from a distressed asset investor. The 60-year-old house was in disrepair from abandonment by the previous owners and infested with rodents and other pests. The house is located 30 miles from Mr. Smith's job as a service man at the airport. His income of approximately \$24,000 per year is about half of what is deemed to be a self-sufficient wage<sup>2</sup> for Merced County. His gas costs were approximately \$30 per week, commuting in his old truck which had been driven for 200,000 miles before he retired it. This case study describes how GRID Alternatives and Valley Clean Air Now (Valley CAN) combine solar rooftop with electric vehicle (EV) incentives to help suburban households like the Smiths.

### 1.2. Suburban definition

No single definition of suburban exists. While many organizations

have attempted to define suburban, a commonly used definition has not been found and no consensus among academics, city planners, and researchers on what constitutes a suburb has been established.<sup>3</sup> Some analysts have tried to use census data to define suburban, making “a rough approximation by taking the urban (metropolitan) area and subtracting census-defined central city municipalities. This approach works better in areas with many local governments than in places like Texas where annexations mean that many central cities take up much of the metropolitan area.”<sup>4</sup>

It is important to note that “suburbs are actually extremely heterogeneous in their demographic composition, economy, and land use patterns.”<sup>5</sup> Yet, several characteristics more frequently found in suburbs contribute to the following key challenges unique to low- and moderate-income (LMI) communities in suburban areas.

### 1.3. Suburban challenges

#### 1.3.1. Rising housing costs during flat wage growth

Suburbs are usually thought of as affluent neighborhoods; however, a study by the Brookings Institution found that “the suburban poor outnumbered their city counterparts by almost one million people in 2005. During the first year of the recession that began in 2007, suburbs

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<sup>1</sup> The Smiths family name has been changed to protect their privacy.

<sup>2</sup> Self-Sufficiency Standard, available at: <http://www.selfsufficiencystandard.org/self-sufficiency-standard-0>.

<sup>3</sup> Ann Forsyth. 2012. Defining Suburbs. Journal of Planning Literature 27, no. 3: 270–281. [https://dash.harvard.edu/bitstream/handle/1/16139611/Forsyth\\_Defining\\_Suburbs2012.pdf?sequence=1](https://dash.harvard.edu/bitstream/handle/1/16139611/Forsyth_Defining_Suburbs2012.pdf?sequence=1).

<sup>4</sup> Ann Forsyth. 2012. Defining Suburbs. Journal of Planning Literature 27, no. 3: 270–281. [https://dash.harvard.edu/bitstream/handle/1/16139611/Forsyth\\_Defining\\_Suburbs2012.pdf?sequence=1](https://dash.harvard.edu/bitstream/handle/1/16139611/Forsyth_Defining_Suburbs2012.pdf?sequence=1).

<sup>5</sup> Alexandra Murphy and Scott W. Allard, The changing geography of poverty, Focus Vol. 32, No. 1, Spring/Summer 2015.

added more than twice as many poor people as did their cities.”<sup>6</sup> Some suburbs are still recovering from the financial downturn.<sup>7</sup> In the suburbs, low-income households “are much more likely to be homeowners than the urban poor, reflecting the fact that suburbs were originally designed for homeownership and zoning practices have reinforced those intentions.”<sup>8</sup> As housing costs have continued to rise while wages have remained mostly flat, suburban poor homeowners struggle financially to remain in their homes and provide for maintenance and upkeep of their houses.

### 1.3.2. Insufficient public transportation options

Although some suburbs are dense and close to the city center, other suburbs are geographically spread out. Increasingly, as housing costs rise, people have been forced to move further and further from work. Between 2000 and 2017, jobs within a typical commuter’s distance to suburbs fell by 7%.<sup>9</sup> Suburbs often do not have sufficient public transportation options. This forces LMI communities to rely on motor vehicles for commuting needs, resulting in an increased share of transportation costs of the household income.<sup>10</sup> In fact, lower-income families spend a much larger percentage of their income on transportation cost than upper income households. In 2014 8.2% of household income was spent on transportation costs by the upper third income bracket compared to 15.7% by the lower third, almost double the amount of the upper third.<sup>11</sup>

### 1.3.3. Access to safety net and support

Adding to the need to commute far distances for work, support services for LMI families are often located in city centers. Overall, suburban areas are much more likely to lack organizations that provide shelter and food assistance, employment organizations, and education organizations, which are key to upward mobility.<sup>12</sup> Additionally, suburban anti-poverty organizations often cover geographically larger areas, making outreach financially and logistically challenging. LMI communities in the suburbs thus have less assistance available nearby and therefore need to spend more money and/or time on transportation to be able to access these services in a nearby urban environment.

## 1.4. Economic background

The case study pilot was implemented in the city of Winton in Merced County, Calif., situated in the San Joaquin Valley. In 2010, the

<sup>6</sup> Steven Raphael and Michael Stoll, *Job Sprawl and the Suburbanization of Poverty*, Metropolitan Policy Program at Brookings, March 30, 2010, <https://www.brookings.edu/research/job-sprawl-and-the-suburbanization-of-poverty/>.

<sup>7</sup> Alana Semuels, *The Places that May Never Recover from the Recession*, The Atlantic, Dec. 29, 2017, <https://www.theatlantic.com/business/archive/2017/12/suburban-poverty-and-recession/549350/>.

<sup>8</sup> Alexandra Murphy and Scott W. Allard, *The changing geography of poverty*, Institute for Research on Poverty, Focus Vol. 32, No. 1, Spring/Summer 2015.

<sup>9</sup> Elizabeth Kneebone and Natalie Holmes, *The growing distance between people and jobs in metropolitan America*, Metropolitan Policy Program at Brookings, March 2015, [https://www.brookings.edu/wp-content/uploads/2016/07/Srvy\\_JobsProximity.pdf](https://www.brookings.edu/wp-content/uploads/2016/07/Srvy_JobsProximity.pdf).

<sup>10</sup> According to the Bureau of Labor Statistics, transportation costs are the second largest share of household expenditures, after housing. Depending on the make-up of the family, transportation costs make up between 15.9% and 18.6% of the household expenditures. (BLS, <https://www.bls.gov/news.release/cesan.nr0.htm>).

<sup>11</sup> Pew Charitable Trusts, *Household Expenditures and Income*, March 30, 2016, <http://www.pewtrusts.org/en/research-and-analysis/issue-briefs/2016/03/household-expenditures-and-income>.

<sup>12</sup> Alexandra Murphy and Scott W. Allard, *The changing geography of poverty*, Institute for Research on Poverty, Focus Vol. 32, No. 1, Spring/Summer 2015.

population in Winton was 10,613.<sup>13</sup> The median household income in the Winton zip code is \$42,705, which is lower than U.S. (\$55,322) and California (\$63,783) income.<sup>14</sup> In Winton, 21.9% of people live in poverty (compared to 15.1% in the U.S.), which is generally consistent with California’s inland valleys, and 8.4% live in deep-poverty (compared to 6.7% in the U.S.), earning less than half of the federal poverty level.<sup>15</sup> Employment in the San Joaquin Valley is largely concentrated in the agricultural sector, with over 19% of the population employed in this season-dependent industry.<sup>16</sup> The unemployment rate is fairly high in Merced County - 10.6%, more than double the national average of 4.1% as of February 2018.<sup>17</sup> The unemployment rate reflects the large role agriculture plays in the region with strong seasonal swings, displaying high winter peaks and lower numbers in the summer and fall.<sup>18</sup>

## 1.5. Housing stock and energy burden

Housing in Merced County, Calif, is almost evenly split between owner-occupied and renter-occupied units; in 2010, 54.5% of housing units were owner-occupied and 45.5% renter-occupied.<sup>19</sup> The majority of the housing stock in the county, over 75%, is detached single-family units, reflecting the suburban environment. Mobile homes represent the second largest amount of housing units at 6.2%.<sup>20</sup> The average family spends approximately \$2,148/year (owners) or \$1,708/year (renters) on home energy costs. This translates to an energy burden of almost 16% for homeowners and approximately 13% for renters that make 0–30% of Area Median Income (AMI) (see Fig. 1<sup>21,22</sup>). The calculations for energy burden do not include transportation costs, only electricity and heating fuel.<sup>23</sup>

## 1.6. Transportation costs

Transportation costs for Winton are higher across different fuel sources compared to the national average. Gasoline costs \$2.84/gallon (compared to the U.S. national average of \$2.38/gallon) and diesel costs \$2.89/gallon (compared to \$2.55/gallon).<sup>24</sup> However, the fuel costs are slightly lower than the average fuel costs in California, where gasoline is \$3.08/gallon and diesel \$3.07/gallon in 2017.<sup>25</sup> In addition, the suburban region is highly car-dependent – more so than the U.S. national average. Some 80% of work commuters in the relevant zip

<sup>13</sup> 2010 Census Interactive Population Search, <https://www.census.gov/2010census/popmap/ipmtxt.php?fl=06:0686076>. (Accessed 30 March 2018).

<sup>14</sup> TownCharts, <http://www.towncharts.com/California/Economy/95388-Zipcode-CA-Economy-data.html>. (Accessed 30 March 2018).

<sup>15</sup> Headwater Economics, *Populations at Risk*, <https://headwaterseconomics.org/tools/populations-at-risk/>. (Accessed on 9 February 2018).

<sup>16</sup> TownCharts, <http://www.towncharts.com/California/Economy/95388-Zipcode-CA-Economy-data.html>. (Accessed 30 March 2018).

<sup>17</sup> Bureau of Labor Statistics, *Local Area Unemployment Statistics*. (Accessed 9 May 2018).

<sup>18</sup> Bureau of Labor Statistics, *Local Area Unemployment Statistics*. (Accessed 9 May 2018).

<sup>19</sup> Census, *General Housing Characteristics*, Merced County, 2010.

<sup>20</sup> American Community Survey, Census, *General Housing Characteristics*, Merced County, 2016.

<sup>21</sup> DOE’s Low Income Energy Affordability Data (LEAD) tool, <https://openei.org/doe-opendata/dataset/celica-data>. (Accessed on 9 February 2018).

<sup>22</sup> The graph contains income levels that follow HUD guidelines: 0–30% Area Median Income (AMI) – extremely low income, 30–50% AMI – very low income, 50–80% AMI low income, Over 80% AMI is no longer considered low income, and 100%+ AMI is above median income.

<sup>23</sup> For more information on the methodology, please see <https://openei.org/doe-opendata/dataset/celica-data>.

<sup>24</sup> DOE’s EERE State and Local Energy Data tool, <https://apps1.eere.energy.gov/sled/>. (Accessed 9 February 2018).

<sup>25</sup> EIA, *Weekly Retail Gasoline and Diesel Prices*, [https://www.eia.gov/dnav/pet/pri\\_gnd\\_dcus\\_sca\\_a.htm](https://www.eia.gov/dnav/pet/pri_gnd_dcus_sca_a.htm). (Accessed 17 May 2018).

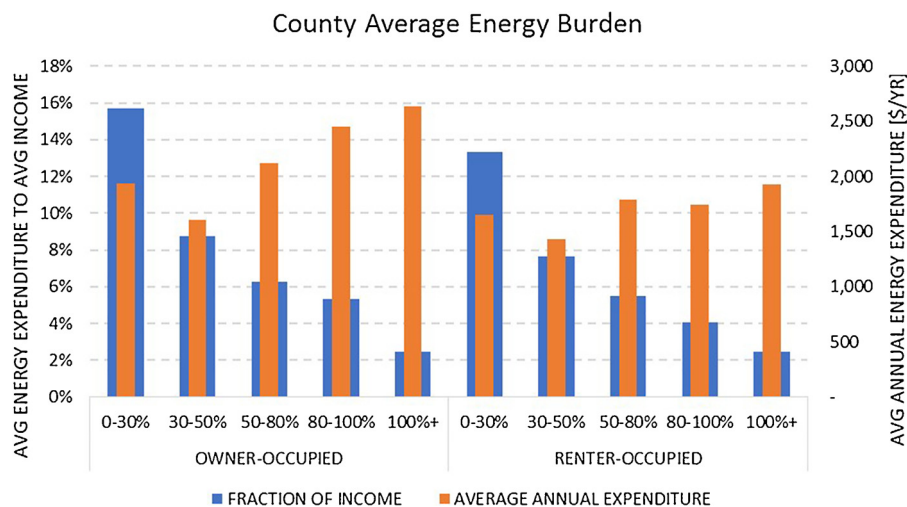


Fig. 1. Merced County average energy burden.

Source: U.S. Department of Energy, Low Income Energy Affordability Data (LEAD) tool, <https://openei.org/doe-opendata/dataset/celica-data>.

code drove to work, followed by other vehicle-dependent modes of transportation such as carpooling, taxicab, or motorcycle.<sup>26</sup> Only 6% of the people in Winton indicated that they work and live in the same city, which is much lower than the national average of 42% and the average in California of 37%.<sup>27</sup> This makes commuting costs a much more important part of household expenditures. According to the Center for Neighborhood Technology's Housing and Transportation Affordability Index, households in Winton spend 38% of their household income on transportation. Together with 37% on housing, the total combined housing and transportation costs are 75%.<sup>28</sup>



Fig. 2. Suburban neighborhood in Central Valley.

Source: U.S. Department of Energy field study, 2018.

## 2. Valley Clean Air Now and GRID Alternatives EV Program and customer case example

### 2.1. Customer case experience

As introduced at the beginning of the article, this case study examines the case of the Smiths, who live in a 60-year-old house, have a household income of \$24,000 and rely on a truck with 200,000 miles on it. Fig. 2<sup>29</sup> is a representative photo of their neighborhood.

Today, the family lives in a home that has been stripped down to the studs to remove mold and create a more livable home. Mr. Smith does this construction himself as his finances allow. After hearing about Valley CAN from friends, Mr. Smith attended one of its Tune In & Tune Up events to learn about his options to either repair or replace his car. Valley CAN assisted Mr. Smith with the qualification process for the Enhanced Fleet Modernization Program Plus-Up (EFMP Plus-Up), which took about three weeks. Mr. Smith qualified for an EFMP Plus-Up down payment incentive of \$9,500 toward a battery electric vehicle (BEV), and Mr. Smith paid the remaining cost (about \$2,500) of the vehicle purchase.

EFMP Plus-Up customers additionally qualify for a \$2,000 incentive towards the purchase and installation of a home EV charger. One barrier to claiming this incentive is that many LMI homes need an upgrade to their home service equipment in order to add a circuit for the charger. To work

around this barrier, by combining available incentives, Valley CAN referred Mr. Smith to GRID Alternatives (GRID). Approximately seven months later, it contacted Mr. Smith to participate in the Single Family Affordable Solar Housing (SASH) program, which includes funding for needed home service equipment upgrades. After a thorough assessment, GRID installed a 4.56 kW solar PV system and a Level 2-EV charger for Mr. Smith's family. The project was connected to the utility in August 2017. Through the third-party ownership model coordinated by GRID, which includes a 20-year contract with SunRun, Mr. Smith's costs were defrayed to zero.

GRID leveraged the following programmatic funds for this project (Fig. 3<sup>30</sup>):

- California's Low-Income Weatherization Program (LIWP)
- California's Enhanced Fleet Modernization Program (EFMP) Plus Up
- SunRun third-party ownership (TPO) partnership to monetize the federal solar investment tax credit (ITC)

### 2.2. Program coordinator

The program coordinator, GRID Alternatives, was founded in 2001 and has been administering the SASH Program in California since 2009.<sup>31</sup> The non-profit provides no- to very-low-cost solar power for

<sup>26</sup> TownCharts, <http://www.towncharts.com/California/Economy/95388-Zipcode-CA-Economy-data.html>. (Accessed 30 March 2018).

<sup>27</sup> TownCharts, <http://www.towncharts.com/California/Economy/95388-Zipcode-CA-Economy-data.html>. (Accessed 30 March 2018).

<sup>28</sup> H&T Affordability Index, <https://htaindex.cnt.org/map/>. (Accessed 4 June 2018).

<sup>29</sup> U.S. Department of Energy field study, 2018.

<sup>30</sup> Graphic representation of program components by U.S. Department of Energy, 2018.

<sup>31</sup> The Single-family Affordable Solar Housing (SAHS) Program supports low-income homeowners to offset the costs of a solar system. For more information, please see <http://www.gosolarcalifornia.ca.gov/affordable/sash.php>.

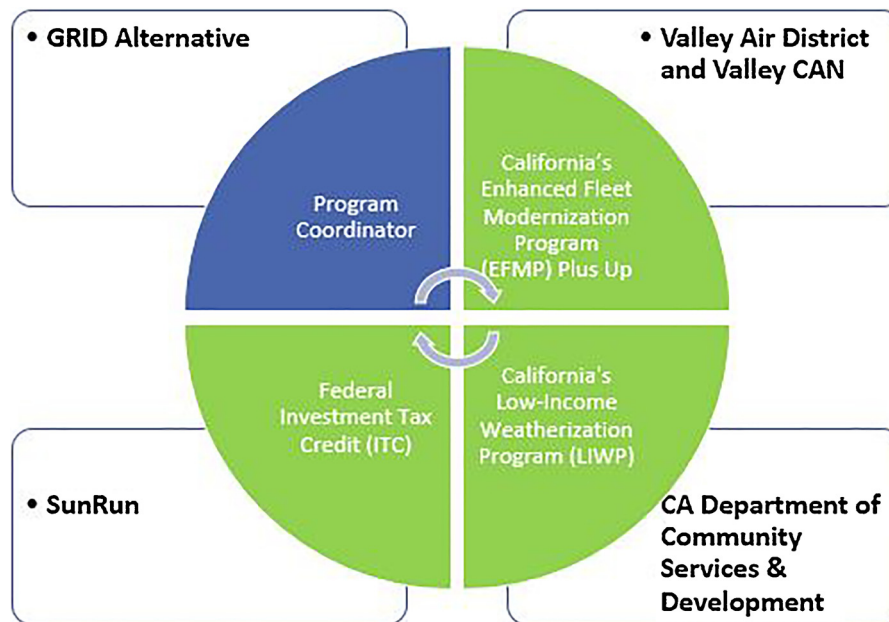


Fig. 3. GRID's EV pilot program components.  
Source: Author analysis, 2018.

low-income families while enabling hands-on installation experience for job seekers and community volunteers. They also offer technical assistance and turnkey installation services to multifamily affordable housing developers. GRID is working with cooperative, municipal, and investor-owned utilities to develop community solar projects dedicated to low-income communities.

### 2.3. Capital providers

#### 2.3.1. California's EFMP Plus Up programs

For the pilot program, GRID partnered with ValleyCAN, which leveraged California's EFMP Plus Up program funded by the California Air Resources Board. The EFMP Plus Up Program receives funding from the California carbon cap-and-trade program, and uses that funding to provide up to \$9,500 down payment assistance for low-income customers to replace their high-emitting older vehicle with a used plug-in vehicle. The program also offers an additional \$2,000 toward purchase and installation of EV chargers for customers who choose a BEV.

Valley CAN is a 501(c)(3) public charity committed to quantifiably improving air quality in California's San Joaquin Valley; a region with some of the worst air quality in the nation. The non-profit offers the EFMP Plus Up Program for residents of the San Joaquin Air Valley Air District whose income is at or below 400% poverty level. Their vehicle must be currently operational, recently failed a smog/emissions test, gas-powered, and a 1999 model year or older.<sup>32</sup> The program then offers incentives for customers to replace their vehicle with a hybrid or plug-in hybrid electric vehicle (PHEV).<sup>33</sup>

For projects where GRID cannot access the EFMP Plus Up \$2,000 grants, GRID works to identify other means to finance EV charger installation, including in-kind donations from equipment providers, local

grants, and opportunities to find cost efficiencies within the solar installation process.

#### 2.3.2. LIWP

Funding for the solar installation for the Smith's pilot project also came from the state's Low-Income Weatherization Program (LIWP). California's Department of Community Services & Development (CSD) "administers [LIWP] funded by Cap-and-Trade auction proceeds and one of California's Climate Investments, and the Drought Emergency Assistance Program."<sup>34</sup> LIWP installs solar photovoltaics (PV), solar hot water heaters, and energy efficiency measures in eligible low-income single-family and multi-family dwellings in disadvantaged communities to reduce GHG emissions and save energy.<sup>35</sup> California uses the CalEnviro Screen tool<sup>36</sup> to target the highest scoring 25% of census tracts (the higher the score the more pollution the community experiences).<sup>37</sup>

Low-income communities are defined as "census tracts that are either at or below 80% of the statewide median income, or at or below the threshold designated as low-income by the California Department of Housing and Community Development's (HCD) 2016 State Income Limits."<sup>38</sup> In 2012, Senate Bill 535 directed "that 25% of the proceeds from the Greenhouse Gas Reduction Fund go to projects that provide a benefit to disadvantaged

<sup>34</sup> California Department of Community Services & Development, Partners Against Poverty, <http://www.csd.ca.gov/AboutUs.aspx>. (Accessed 30 March 2018).

<sup>35</sup> California Department of Community Services & Development, Low-Income Weatherization Program, <http://www.csd.ca.gov/liwp.aspx>. (Accessed 30 March 2018).

<sup>36</sup> Office of Environmental Health Hazard Assessment (OEHA), CalEnviro Screen, <https://oehha.ca.gov/calenviroscreenhttps://oehha.maps.arcgis.com/apps/webappviewer/index.html?id=4560cfbce7c745c299b2d0cbb07044f5> accessed May 9, 2018. (Accessed 9 May 2018).

<sup>37</sup> California OEHA, SB 535 Disadvantaged Communities, <https://oehha.ca.gov/calenviroscreen/sb535>. (Accessed 24 May 2018).

<sup>38</sup> California Air Resources Board, Disadvantaged and Low-income Communities Investment, <https://www.arb.ca.gov/cc/capandtrade/auctionproceeds/communityinvestments.htm>. (Accessed 30 March 2018).

<sup>32</sup> Valley CAN, Vehicle Replacement Eligibility, <https://valleycan.org/vehicle-replacement/>. (Accessed 30 March 2018).

<sup>33</sup> California Air Resources Board, Enhanced Fleet Modernization Program – Car Scrap, <https://www.arb.ca.gov/msprog/aqip/efmp/efmp.htm>. (Accessed 30 March 2018).

|                                                 | Electricity | Total          |
|-------------------------------------------------|-------------|----------------|
| Electricity Bill per Year (BEFORE)              | \$2,400     |                |
| Gas Costs per Year (BEFORE)                     | \$2,009     |                |
| <b>Total Costs per Year (BEFORE)</b>            |             | <b>\$4,409</b> |
| Electricity Bill per Year (AFTER)<br>– low-end  | \$240*      |                |
| EV Charging (AFTER)                             | \$674       |                |
| Electricity Bill per Year (AFTER)<br>– high-end | \$914**     |                |
| <b>Total Savings – low-end</b>                  |             | <b>\$3,495</b> |
| <b>Total Savings – high-end</b>                 |             | <b>\$4,169</b> |

\*estimate prior to year-end Plus up

\*\*assumes cost of EV charging is in addition to \$20/ month estimate given by customer.

Fig. 4. Analysis on the economics of using rooftop solar to offset gasoline use.

\*estimate prior to year-end Plus up.

\*\*assumes cost of EV charging is in addition to \$20/ month estimate given by customer.

Source: Author analysis, 2018.

communities.”<sup>39</sup> In 2016, AB 1559 required that “25% of proceeds from the fund be spent on projects located in disadvantaged communities”<sup>40,41</sup>.

The eligibility of LIWP is generally 80% of AMI; however, in some counties 60% of SMI is used (see Appendix A).

The LIWP requires annual allocation in the California state budget from cap-and-trade proceeds.<sup>42</sup> While the LIWP was initially funded with \$75 million and \$78.8 million in the 2014/2015 California state budget and the 2015/2016 budget respectively, the funding has dropped to \$18 million in the 2017/2018 budget and is projected to be zeroed out in the 2018/2019 budget. Additionally, each allocation comes with time limits to spend the funding, making it difficult to embark on new programs. The program has installed solar systems in about 2,900 homes to date.

### 2.3.3. Sunrun third-party ownership partnership

GRID partnered with Sunrun to monetize the investment tax credit (ITC) and provide this benefit to low-income customers. Through this partnership, Sunrun serves as a third-party owner (TPO) and owns, operates, maintains, and insures the solar systems that GRID installs. GRID funds each customer's PPA or solar lease for 20 years. Through this model, Sunrun is able to leverage the federal investment tax credit and depreciation benefits, which reduces the overall cost of the solar installation for the customer.<sup>43</sup> The TPO model also allows GRID to overcome the gap financing, which remains one of the primary obstacles of LMI customers to participate in GRID's programs. While SASH funding and other incentives provide support to LMI customers interested in solar installations, a gap between funding and the cost of the system usually remains because LMI customers are often not able to assume more debt.



Fig. 5. Exterior of Mr. Smith's home with 2 cars.

Source: U.S. Department of Energy field study, 2018.

GRID closes this gap via a grant financed by in-kind donations, corporate sponsorships, and general philanthropy.<sup>44</sup> Additionally, the TPO model provides GRID the ability to overcome the gap in financing, while providing warranty coverages and system monitoring. TPO played a role in over 70% of Completed SASH 2.0 Projects.<sup>45</sup>

## 3. Results and discussion

### 3.1. Customer savings

Prior to the installation of the solar panels, Mr. Smith was paying at least \$220 per month in electricity costs, including electricity for heating purposes. Due to poor insulation, the use of space heaters drove up the costs. After installation of the solar panels, Mr. Smith now pays approximately \$20 per month for heat, electricity, and charging his EV every night. Due to the limited range of the EV, it is primarily used as a

<sup>39</sup> California OEHHA, Disadvantaged Community Designation (April 2017), <https://oehha.ca.gov/calenviroscreen/sb535>.

<sup>40</sup> California OEHHA, Disadvantaged Community Designation (April 2017), <https://oehha.ca.gov/calenviroscreen/sb535>.

<sup>41</sup> California Air Resources Board, Disadvantaged and Low-income Communities Investment, <https://www.arb.ca.gov/cc/capandtrade/auctionproceeds/communityinvestments.htm> accessed (Accessed 30 March 2018).

<sup>42</sup> Isaac Sevier, California's Best Low-Income Climate Program is in Jeopardy, Natural Resources Defense Council, <https://www.nrdc.org/experts/isaac-sevier/californias-best-low-income-climate-program-jeopardy>, (Accessed 24 May 2018).

<sup>43</sup> Sunrun, Press Release - Sunrun Partners with GRID Alternatives to Finance Solar Projects for Low-Income Homeowners, April 25, 2017, <http://investors.sunrun.com/news-releases/news-release-details/sunrun-partners-grid-alternatives-finance-solar-projects-low>.

<sup>44</sup> GRID Alternatives, Single-family Affordable Solar Homes (SAHS Program) – Semi-Annual Program Status Report, January 2018, [https://gridalternatives.org/sites/default/files/FINAL\\_SASH\\_Semi%20annual%20report\\_Q3Q4\\_1.30.2018.pdf](https://gridalternatives.org/sites/default/files/FINAL_SASH_Semi%20annual%20report_Q3Q4_1.30.2018.pdf).

<sup>45</sup> GRID Alternatives, Single-family Affordable Solar Homes (SAHS Program) – Semi-Annual Program Status Report, January 2018, [https://gridalternatives.org/sites/default/files/FINAL\\_SASH\\_Semi%20annual%20report\\_Q3Q4\\_1.30.2018.pdf](https://gridalternatives.org/sites/default/files/FINAL_SASH_Semi%20annual%20report_Q3Q4_1.30.2018.pdf).

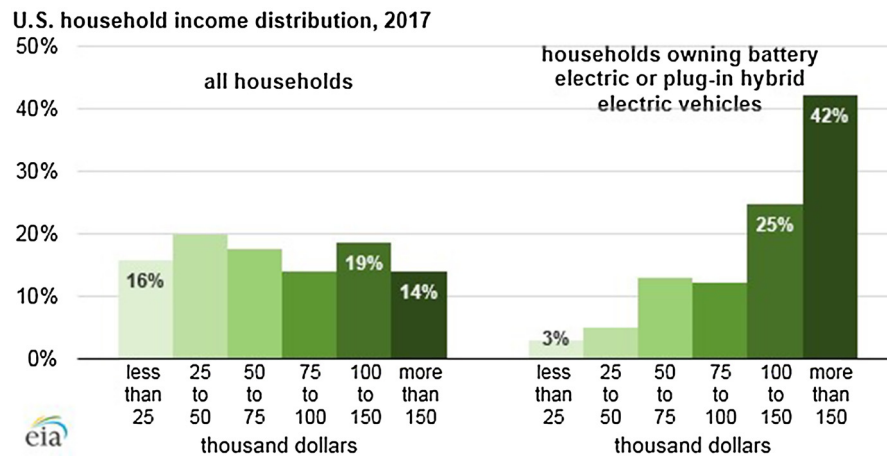


Fig. 6. U.S. household income distribution for all households compared to households with battery electric or plug-in hybrid electric vehicles, 2017. Source: U.S. Energy Information Administration, U.S. Department of Transportation National Household Travel Survey, 2017.

second car, but this is a benefit that his family may not otherwise have been able to afford. By using a newly purchased EV, Mr. Smith saves not only on gas, but also on insurance costs since this is a newer car with fewer risks. In addition, the full upfront payment avoided potential predatory car lending practices. A preliminary calculation shows that using solar PV to offset gasoline could lead to monetary savings between 79%–95% of his original electricity and gas payments (Figs. 4<sup>46</sup> and 5<sup>47</sup>).

### 3.2. Lessons learned

Deploying solar and EV interventions at the same time is a holistic approach to LMI consumers' energy needs. California made effective use of their data to create and leverage the CalEnviroScreen to ensure that scarce resources were targeted to communities in need and developed a variety of innovative policies to encourage EV adoption in low-income communities. However, a practitioner gap on the ground remains in terms of supporting successful deployment of those policies.

Many low-income families have outdated electrical services in their homes that cannot accommodate the additional circuit for an EV charger. This is an issue GRID is already addressing in its solar program through main service panel upgrades. Additionally, in some cases with vehicles that contain small batteries, GRID is able to combine both interventions without installing a charger at all. A "Level 1" charging of the vehicle through a standard home electrical outlet is sufficient in those cases.

The key challenge is the coordination of different low-income EV programs and services. They are run by different entities, have different names and eligibility criteria, and market themselves in different ways. In combination with a novel underlying technology that low-income communities lack exposure to, it can be difficult to coordinate those programs, especially at scale. Outreach is especially challenging if there are many different programs that compete for the time and attention of LMI consumers. The cooperation between GRID and Valley CAN is addressing the outreach challenge by providing a seamless experience to the customer in an innovative way.

## 4. Conclusion

This case study highlights the effective collaboration of various sources of funding to create a low-income appropriate approach to leveraging incentives for both rooftop solar and EVs. Together they

made a significant economic impact on the customer's monthly electricity and transportation costs.

This case also highlights the fact that although low-income individuals could potentially benefit the most from solar, EV, and battery savings, they often have the least access to it. Most solar programs have not been appropriately structured for the low-income market. For example, Sunrun's participation was needed in this case to monetize the tax equity that would otherwise not be available to a low-income customer with few tax assets. In addition, funding from California's carbon market mechanism, which is not available in most states, made grant funding for this pilot possible. In many other states, low-income ratepayers and taxpayers often pay into the funding pools for rebates and incentives for solar. Lack of upfront capital or credit scores, however, prevent them from reaping the benefits of these programs. For example, all ratepayers in Colorado pay into the Renewable Energy Standard Adjustment (RESA) account with 2% of their monthly bill, but low-income customers have not had proportionate, direct access to resultant solar incentives.<sup>48</sup>

EIA reported in 2018 that, although approximately one-third of all American households have annual incomes higher than \$100,000, about two-thirds of households with BEV or PHEV have incomes higher than \$100,000 (Fig. 6<sup>49</sup>). In addition, households with annual incomes lower than \$25,000 account for only 3% of BEV- and PHEV-owning households but make up 16% of all households in the U.S.

If the estimates from EIA are indicative of a trend of limited accessibility to EV technology by higher income households then the trajectory of this trend could be mitigated by dedicating larger per capita awards exclusively for the benefit of low-income customers. Building on recent pilot programs, such as the Solar in Your Community Challenge, a new LMI Solar program could be dedicated to low-income solar incentives. The program could cover the costs of rooftop solar for qualifying families, particularly in states and regions with ample solar technical potential.<sup>50</sup> According to the National Renewable Energy Laboratory (NREL), annual residential solar potential is approximately

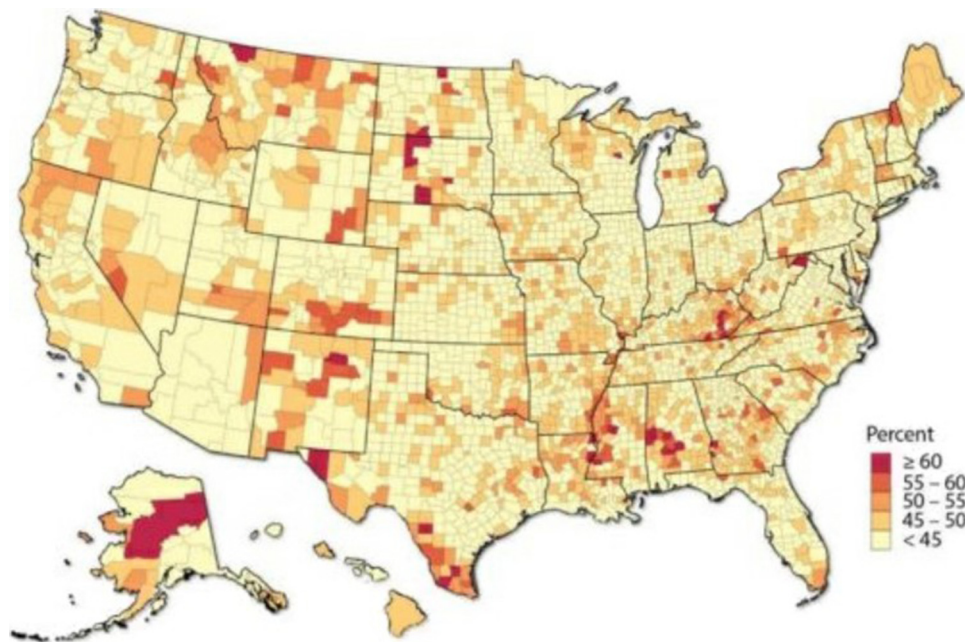
<sup>48</sup> Opening Testimony of Tom Figel for GRID, Proceeding Nos. 16AL-0048E, 16A-0055E, 16A-0139E, Before the public utilities commission of the State of Colorado.

<sup>49</sup> U.S. Energy Information Administration, U.S. Department of Transportation National Household Travel Survey, 2017.

<sup>50</sup> DSIRE, NC Clean Energy Technology Center, [http://ncsolarcen-prod.s3.amazonaws.com/wp-content/uploads/2017/11/DSIRE\\_Net\\_Metering\\_November2017.pdf](http://ncsolarcen-prod.s3.amazonaws.com/wp-content/uploads/2017/11/DSIRE_Net_Metering_November2017.pdf). (Accessed 2 April 2018).

<sup>46</sup> Author Analysis, 2018.

<sup>47</sup> U.S. Department of Energy field study, 2018.



**Fig. 7.** County LMI rooftop technical potential as percent of total residential potential.

Source: Ben Sigrin and Meghan Mooney, 2018. Rooftop Solar Technical Potential for Low-to-Moderate Income Households in the United States. Golden, CO: National Renewable Energy Laboratory. NREL/TP-6A20-70901. <https://www.nrel.gov/docs/fy18osti/70901.pdf>.

1,000 TWh (terawatt-hours), where the LMI opportunity is 415.9 TWh, nearly half (42%) of total annual residential solar potential (Fig. 7<sup>51</sup>).<sup>52</sup>

The creation of a low-income-specific funding mechanism like the EV pilot program in California could provide more balanced access to EV technologies by various income groups. Such a program could meet the structural barriers inherent to reaching low-income populations such as lack of available upfront capital for participation. The program could instead provide grants to pay for upfront costs. An LMI PEV Grant Program could ensure that low-income customers will have equal access to the savings afforded by these technologies.

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<sup>51</sup> NREL, 2018, [maps.nrel.gov/solar-for-all](https://maps.nrel.gov/solar-for-all). (Accessed 25 May 2018).

<sup>52</sup> Sigrin, Ben and Meghan Mooney. 2018. Rooftop Solar Technical Potential for Low-to-Moderate Income Households in the United States. Golden, CO: National Renewable Energy Laboratory. NREL/TP-6A20-70901. <https://www.nrel.gov/docs/fy18osti/70901.pdf>.

## Appendix A

## California 2018 AMI-2017 SMI Income Guidelines

Source: Low-Income Weatherization Program (LIWP), Department of Community Services and Development, 2018.

| FY 2017 60% SMI Guidelines <sup>2</sup> → | \$25,103                                                                                               | \$32,827 | \$40,551 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$69,516  | \$70,964  | \$72,413  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|                                           | HCD FY2018 80% AREA MEDIAN INCOME, BY COUNTY <sup>1</sup> (except at 60% SMI level, where highlighted) |          |          |           |           |           |           |           |           |           |           |           |
| No. in Household →<br>County ↓            | 1                                                                                                      | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        |
| Alameda County                            | \$56,300                                                                                               | \$64,350 | \$72,400 | \$80,400  | \$86,850  | \$93,300  | \$99,700  | \$106,150 | \$112,550 | \$119,000 | \$125,400 | \$131,850 |
| Alpine County                             | \$46,100                                                                                               | \$52,650 | \$59,250 | \$65,800  | \$71,100  | \$76,350  | \$81,600  | \$86,900  | \$92,100  | \$97,400  | \$102,650 | \$107,900 |
| Amador County                             | \$40,500                                                                                               | \$46,300 | \$52,100 | \$57,850  | \$62,500  | \$67,150  | \$71,750  | \$76,400  | \$81,000  | \$85,600  | \$90,250  | \$94,850  |
| Butte County                              | \$35,100                                                                                               | \$40,100 | \$45,100 | \$50,100  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$70,150  | \$74,150  | \$78,150  | \$82,150  |
| Calaveras County                          | \$39,350                                                                                               | \$44,950 | \$50,550 | \$56,150  | \$60,650  | \$65,150  | \$69,650  | \$74,150  | \$78,600  | \$83,100  | \$87,600  | \$92,100  |
| Colusa County                             | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Contra Costa County                       | \$56,300                                                                                               | \$64,350 | \$72,400 | \$80,400  | \$86,850  | \$93,300  | \$99,700  | \$106,150 | \$112,550 | \$119,000 | \$125,400 | \$131,850 |
| Del Norte County                          | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| El Dorado County                          | \$42,650                                                                                               | \$48,750 | \$54,850 | \$60,900  | \$66,800  | \$70,650  | \$75,550  | \$80,400  | \$85,250  | \$90,150  | \$95,000  | \$99,900  |
| Fresno County                             | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Glenn County                              | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Humboldt County                           | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Imperial County                           | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Inyo County                               | \$40,350                                                                                               | \$46,100 | \$51,850 | \$57,600  | \$62,250  | \$66,850  | \$71,450  | \$76,050  | \$80,650  | \$85,250  | \$89,850  | \$94,450  |
| Kern County                               | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Kings County                              | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Lake County                               | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Lassen County                             | \$38,850                                                                                               | \$44,400 | \$49,950 | \$55,500  | \$59,950  | \$64,400  | \$68,850  | \$73,300  | \$77,700  | \$82,150  | \$86,600  | \$91,000  |
| Los Angeles County                        | \$50,500                                                                                               | \$57,700 | \$64,900 | \$72,100  | \$77,900  | \$83,650  | \$89,450  | \$95,200  | \$100,950 | \$106,700 | \$112,500 | \$118,250 |
| Madera County                             | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Marin County                              | \$73,750                                                                                               | \$84,300 | \$94,850 | \$105,350 | \$113,800 | \$122,250 | \$130,650 | \$139,100 | \$147,500 | \$155,900 | \$164,350 | \$172,750 |
| Mariposa County                           | \$36,700                                                                                               | \$41,950 | \$47,200 | \$52,400  | \$56,600  | \$63,723  | \$65,171  | \$69,200  | \$73,350  | \$77,550  | \$81,750  | \$85,950  |
| Mendocino County                          | \$34,100                                                                                               | \$39,000 | \$43,850 | \$48,700  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,200  | \$72,100  | \$75,950  | \$79,850  |
| Merced County                             | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Modoc County                              | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Mono County                               | \$44,750                                                                                               | \$51,150 | \$57,550 | \$63,900  | \$69,050  | \$74,150  | \$79,250  | \$84,350  | \$89,450  | \$94,550  | \$99,700  | \$104,800 |
| Monterey County                           | \$45,600                                                                                               | \$52,100 | \$58,600 | \$65,100  | \$70,350  | \$75,550  | \$80,750  | \$85,950  | \$91,150  | \$96,350  | \$101,550 | \$106,750 |
| Napa County                               | \$52,150                                                                                               | \$59,600 | \$67,050 | \$74,500  | \$80,500  | \$86,450  | \$92,400  | \$98,350  | \$104,300 | \$110,250 | \$116,200 | \$122,200 |
| Nevada County                             | \$42,950                                                                                               | \$49,050 | \$55,200 | \$61,300  | \$66,250  | \$71,150  | \$76,050  | \$80,950  | \$85,800  | \$90,700  | \$95,650  | \$100,550 |
| Orange County                             | \$58,450                                                                                               | \$66,800 | \$75,150 | \$83,450  | \$90,150  | \$96,850  | \$103,550 | \$110,200 | \$116,850 | \$123,500 | \$130,200 | \$136,850 |
| Placer County                             | \$42,650                                                                                               | \$48,750 | \$54,850 | \$60,900  | \$66,800  | \$70,650  | \$75,550  | \$80,400  | \$85,250  | \$90,150  | \$95,000  | \$99,900  |
| Plumas County                             | \$35,100                                                                                               | \$40,100 | \$45,100 | \$50,100  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$70,150  | \$74,150  | \$78,150  | \$82,150  |
| Riverside County                          | \$37,550                                                                                               | \$42,900 | \$48,250 | \$53,600  | \$57,900  | \$63,723  | \$66,500  | \$70,800  | \$75,050  | \$79,350  | \$83,600  | \$87,900  |
| Sacramento County                         | \$42,650                                                                                               | \$48,750 | \$54,850 | \$60,900  | \$66,800  | \$70,650  | \$75,550  | \$80,400  | \$85,250  | \$90,150  | \$95,000  | \$99,900  |
| San Benito County                         | \$46,850                                                                                               | \$53,550 | \$60,250 | \$66,900  | \$72,300  | \$77,650  | \$83,000  | \$88,350  | \$93,650  | \$99,000  | \$104,350 | \$109,700 |
| San Bernardino County                     | \$37,550                                                                                               | \$42,900 | \$48,250 | \$53,600  | \$57,900  | \$63,723  | \$66,500  | \$70,800  | \$75,050  | \$79,350  | \$83,600  | \$87,900  |
| San Diego County                          | \$50,950                                                                                               | \$58,200 | \$65,500 | \$72,750  | \$78,600  | \$84,400  | \$90,250  | \$96,050  | \$101,850 | \$107,650 | \$113,500 | \$119,300 |
| San Francisco County                      | \$73,750                                                                                               | \$84,300 | \$94,850 | \$105,350 | \$113,800 | \$122,250 | \$130,650 | \$139,100 | \$147,500 | \$155,900 | \$164,350 | \$172,750 |
| San Joaquin County                        | \$37,150                                                                                               | \$42,450 | \$47,750 | \$53,050  | \$57,300  | \$63,723  | \$65,800  | \$70,050  | \$74,250  | \$78,500  | \$82,750  | \$87,000  |
| San Luis Obispo County                    | \$45,700                                                                                               | \$52,300 | \$58,850 | \$65,350  | \$70,600  | \$75,850  | \$81,050  | \$86,300  | \$91,500  | \$96,700  | \$101,950 | \$107,150 |
| San Mateo County                          | \$73,750                                                                                               | \$84,300 | \$94,850 | \$105,350 | \$113,800 | \$122,250 | \$130,650 | \$139,100 | \$147,500 | \$155,900 | \$164,350 | \$172,750 |
| Santa Barbara County                      | \$50,450                                                                                               | \$57,650 | \$64,850 | \$72,050  | \$77,850  | \$83,600  | \$89,350  | \$95,150  | \$100,850 | \$106,650 | \$112,400 | \$118,150 |
| Santa Clara County                        | \$59,400                                                                                               | \$67,900 | \$76,400 | \$84,900  | \$91,650  | \$98,450  | \$105,250 | \$112,050 | \$118,850 | \$125,650 | \$132,450 | \$139,250 |
| Santa Cruz County                         | \$56,500                                                                                               | \$64,550 | \$72,600 | \$80,650  | \$87,150  | \$93,600  | \$100,050 | \$106,500 | \$112,900 | \$119,350 | \$125,800 | \$132,250 |
| Shasta County                             | \$34,650                                                                                               | \$39,600 | \$44,550 | \$49,450  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$69,250  | \$73,200  | \$77,150  | \$81,100  |
| Sierra County                             | \$39,400                                                                                               | \$45,000 | \$50,650 | \$56,250  | \$60,750  | \$65,250  | \$69,750  | \$74,250  | \$78,750  | \$83,250  | \$87,750  | \$92,250  |
| Siskiyou County                           | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Solano County                             | \$45,500                                                                                               | \$52,000 | \$58,500 | \$65,000  | \$70,200  | \$75,400  | \$80,600  | \$85,800  | \$91,000  | \$96,200  | \$101,400 | \$106,600 |
| Sonoma County                             | \$49,350                                                                                               | \$56,400 | \$63,450 | \$70,500  | \$76,150  | \$81,800  | \$87,450  | \$93,100  | \$98,700  | \$104,350 | \$110,000 | \$115,600 |
| Stanislaus County                         | \$34,750                                                                                               | \$39,700 | \$44,650 | \$49,600  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$69,450  | \$73,400  | \$77,400  | \$81,350  |
| Sutter County                             | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Tehama County                             | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Trinity County                            | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Tulare County                             | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |
| Tuolumne County                           | \$37,200                                                                                               | \$42,500 | \$47,800 | \$53,100  | \$57,350  | \$63,723  | \$65,850  | \$70,100  | \$74,350  | \$78,600  | \$82,850  | \$87,100  |
| Ventura County                            | \$55,950                                                                                               | \$63,950 | \$71,950 | \$79,900  | \$86,300  | \$92,700  | \$99,100  | \$105,500 | \$111,850 | \$118,250 | \$124,650 | \$131,050 |
| Yolo County                               | \$43,050                                                                                               | \$49,200 | \$55,350 | \$61,500  | \$66,450  | \$71,350  | \$76,300  | \$81,200  | \$86,100  | \$91,000  | \$95,950  | \$100,850 |
| Yuba County                               | \$33,550                                                                                               | \$38,350 | \$43,150 | \$48,275  | \$55,999  | \$63,723  | \$65,171  | \$66,620  | \$68,068  | \$70,900  | \$74,700  | \$78,550  |

<sup>1</sup>The link for the 2017 Area Median Income (AMI) information is: <http://www.hcd.ca.gov/grants-funding/income-limits/state-and-federal-income-limits/docs/inc2k17.pdf><sup>2</sup>FY 2017 60% State Median Income (SMI) information was obtained from the U.S. Office of Community Services' LIHEAP IM 2016-03 State Median Income Estimates for Optional Use in FFY 2016 and Mandatory Use in FFY 2017 published July 5, 2016.

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